Smart Insole: Remote Gait Detection Algorithm Using Pressure Sensors For Toe Walking Rehabilitation

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***Abstract*—Toe walking is a gait pattern without contacting the heel on the ground. Toe walking is observed as an expansion of a normal developmental phase among toddlers. Still, the continuation of toe walking in older children may cause severe conditions on their foot, ankle, and muscle. Gait rehabilitation can be improved by employing a device that identifies gait patterns, and provides users with biofeedback. In this paper, we introduce Smart Insole platform that remotely monitors walking patterns and provides feedback to users regarding their toe walks. Smart Insole platform detects physical activities using two pressure sensors on the custom-designed insole. In regard to provide a feedback to users, a vibration is triggered after successive toe walks. We evaluated the accuracy of our activity recognition algorithm in a study with 10 healthy subjects. Our proposed algorithm detects toe or normal heel-to-toe walking, sitting, and standing with an average accuracy of 91.3%. Our platform provides orthopedic professionals with accurate and high standard protocols to track activity patterns and outgrow toe walking.**

***Index Terms*—activity recognition, gait cycle, wearable device**

I. Introduction

Toe walking is an abnormality of gait defined by the lack of normal heel-to-floor touch during the gait. In other words, it is characterized as the failure to contact the heel with the ground through initial stance stage of the gait cycle and lack of full foot interaction with the floor during the rest of the gait cycle. Toe walking is a common demeanor among children starting to learn to walk. However, most children after the toddler years exceed walking on their tip toes and begin to walk normally with the heel to toe pattern. [1] Toe walking can sometimes be a symptom of certain circumstances, like a short Achilles tendon, muscle dystrophy, cerebral paralysis, and autism spectrum disorder. [2]

Since toe walking applies a high pressure on muscles and the Achilles tendon, children who consistently walk on the balls of their feet may later suffer from foot deformities, ankle movement restrictions, poor alignment of posture, and impaired balance. [2] [3] [4] Depending on the age of the child and whether the child is able to walk flat-footed or not, physicians may go through nonsurgical or surgical treatments for toe walking. The most common nonsurgical treatment is physical therapy, like calf stretching, Achilles tendon stretching, and sit to stand. Placing a splint or brace is another nonsurgical treatment in case when simple stretching does not assist. Botox injection can temporarily loosen the tight calf muscles and let them to stretch with more ease while casting or splinting. However, the sustainability of beneficial impacts of these treatments appears to be brief. [2] [17]

In this paper, we introduce Smart Insole which is a wearable and affordable technology to measure the number of toe and normal steps for remote monitoring of walking patterns. Smart Insole vibrates after consecutive toe steps, or consistently standing or sitting on the tip toe, as a reminder for the patients to heel down their feet. Activity recognition analysis is conducted based on the data receiving from 2 force sensors embedded in the insole. The connected Android application receives and transmits the sensors' values and the activity states to the cloud-based database. Our proposed platform gives the physicians and parents the opportunity to remote monitor patient's activity in a real-time manner and assists the patients in improving their walking patterns.

The remainder of the paper is structured as follows. Section II introduces the related works. The system design of the Smart Insole, walk detection algorithm, Android application and data storage are presented in Section III. Section IV provides the experiments and evaluation of Smart Insole. Finally in Section V, we conclude and outline the future research direction.

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II. Related Works

Multiple researches have been carried out in the past on idiopathic toe walking (ITW) and treatment options. In [6], Williams *et al.* proposed an online 'Toe Walking Tool' questionnaire to encourage the users to distinguish signs and risk factors of toe walking. Their tool displayed a number of exclusionary questions which were developed in a consecutive order. The device would either authorize progress to the next question set or incite the clinician for further medical examinations. Williams *et al.* [7] intended to investigate the literature and distribute a summary of the relationship between toe walking and Sensory Processing Dysfunction(SPD). Engstrm *et al.* [8] explored whether botulinum toxin A (BTX) can enhance the pattern of walking in individuals who toe walk. In [9], Davies *et al.* studied the geology of ITW and the long-term effect of conservative therapy on gait performance. However, none of these treatments have been proven to prevent toe walking.

There have been several studies on wearable systems for human gait analysis in the past. In [10], Xu *et al.* proposed a smart insole for gait analysis system with 48 pressure sensors, 3-axis accelerometer, 3-axis gyroscope, and 3-axis compass. Liu *et al.* [11] researched on a wearable gait analysis using 3-axis accelerometer and gyroscope. Xu *et al.* [12] provided a compressed sensor-based algorithm with a single accelerometer to accurately recognize human activity. Sazonov *et al.* [13] utilized five force sensitive resistors embedded in an insole and a 3-axis accelerometer for common posture and activity recognition of human employing support vector machines. In [14], Carbonaro *et al.* achieved detection of the gait process (heel-strike, stance, heel-off, and swing) by means of built-in force sensors located on the heel and forefoot and a triaxial accelerometer embedded in forefoot, interfacing with the smartphone through WiFi connection. In [15], Hegde *et al.* proposed a pediatric smart shoe system for remote activity (sitting, standing, and walking) detection. They took advantage of five Force Sensitive Resistors (FSR) sensors locating on the heel and toe of an insole and an accelerometer mounted on the insole. In [16], Lin *et al.* presented a smart insole comprises an array of sensors and an IMU for gait analysis. All of these works are useful to track human locomotion and walking status. However, they implemented complicated algorithms and systems for gait analysis. Our proposed systems identifies toe and normal walking, standing, and sitting by integration of only 2 force sensors in an insole. Moreover, Smart Insole has the potential to improve children's unhealthy walking habits. Our affordable smart insole, not only gives effective bio feedbacks to patients to prevent toe walking, but also provides them with accurate activity detection. Furthermore, in our platform the activity detection process is computed on the embedded hardware in the insole and activity states are transmitted to the phone via Bluetooth Low Energy (BLE) connection. Hence there is no phone to insole Bluetooth communication and vibration feedback are provided to the user even with no Bluetooth connectivity.

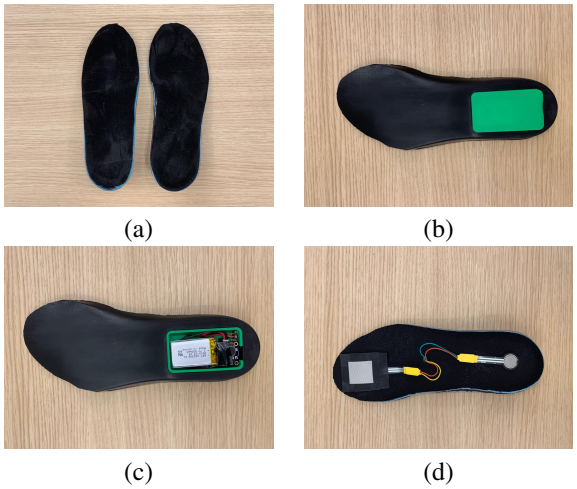

Fig. 1. Data acquisition design components: (a) Final look of insole's design from front. (b) Final look of insole's design from back. (c) Circuit design and board in the insole's container. (d) Sensors' layout on the front surface of insole.

III. Smart Insole

The overall architecture of Smart Insole is composed of three main components. The data acquisition unit collects sensors' readings and sends the time series of sensor measurements to an Android device. Using the collected data, the activity recognition algorithm triggers the mini vibrator embedded in the insole after consecutive toe steps or consistently standing and sitting on the tip toe. The Android application preserves a secure and reliable connection with the data acquisition unit in order to receive the sensors' readings and predicted activities and transmit them to the cloud-based database. The data storage takes advantage of machine learning tools to provide feasible inferences with a visualization.

A. Data acquisition

Physical squeezing and pressure can be detected by FSR [17]. FSR is a resistor, whose value changes depending on the pressure applied to it. In other words, the higher the pressure, the lower the resistance. Resistance changes yield in varying voltage readings at the two ends of the FSR. These voltage values are constantly measured and read by the microcontroller. We utilized Adafruit Feather as the microcontroller board [18], in which 6 analog reading pins are incorporated. The FSRs' voltage readings can be measured by these analog reading pins. Feather is light, tiny, compatible with Arduino, and provides a USB port to program and debug the board. A 4.2/3.7V Lithium Polymer (Lipo/Lipoly) or Lithium Ion (LiIon) battery can be connected to the board through a JST jack provided on the Feather. The battery starts to charge at 100 mA, as soon as a USB cable connects to the Feather. There is a yellow LED on the board that turns on while the battery is charging. The USB port also supplies the power of the Feather board through this time. Feather featured a red LED that blinks while the power supply is on. We inserted an on/off switch on the board to

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reduce the power consumption. Feather benefits from BLE technology which provides transmissions through the board and a phone with the low power, 2.4 GHz spectrum wireless protocol. The sensors' measurements from the analog reading pins are transmitted in real-time via BLE to the Connected Android application.

We used storage sole in which a removable container is housed, and designed our own container to pack the board, a battery, and a mini vibrator in it. Our designed container is visible in Fig1(b). Fig1(c) shows how perfectly our compact system is laid in the container. A square and a round FSR are positioned on the insole's surface which are respectively in contact with toe and heel of foot. Fig1(d) is an illustration of the surface of insole and sensors' location before final fabrication. The FSRs on the surface of the insole are embraced with fabric as in Fig1(a). Since the entire system is integrated in the insole, it can put to use in any shoe. Due to the low cost electronics and storage soles, our proposed data acquisition component is economically efficient. Our objective in developing the system was to minimize cost and power consumption, in addition to patient convenience of use.

B. Activity Recognition Algorithm

The proposed solution algorithm is demonstrated in Fig2 flowchart. After activity recognition, depending on the detected activity, the steps counting or vibration policy procedure takes place.

The measurements of two FSRs located on the toe and heel zone of the insole are received at a frequency of 20Hz via Adafruit BLE in real-time manner. Signals from each FSR are captured and processed by windows of size 8. For each window (t_{th} window) the average and standard deviation are computed by (1) and (2) respectively.

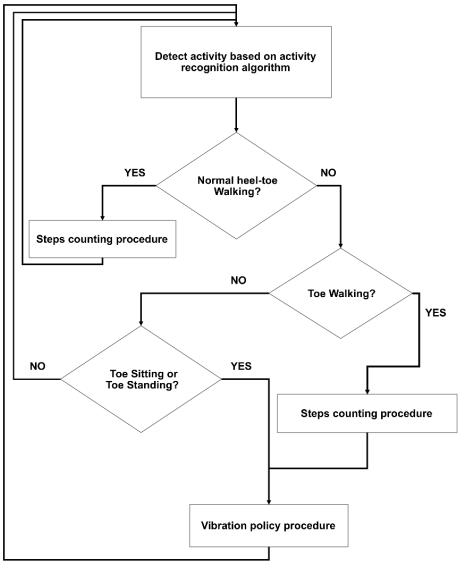

Fig. 2. Proposed algorithm procedure flowchart.

Algorithm 1 Activity Recognition Algorithm

Input: A window containing 8 consecutive instances of the FSR located under the toe (toe window),
A window containing 8 consecutive instances of the FSR located under the heel (heel window);
Output: Activity;
ToeAvr ← Average of a toe window;
HeelAvr ← Average of a heel window;
ToeStd ← Standard deviation of a toe window;
if *ToeStd* > *toeStdWalkDetection* **then**
  if (*ToeAvr* > *toeAvrgToeWalkDetection*)***and*** (*HeelAvr* < *heelAvrgToeWalkDetection*) **then**
    Activity ← Toe Walking;
  else
    Activity ← Normal Walking;
  end
else
  if *ToeAvr* > *toeAvrgToeSitStandDetection* **then**
    if *HeelAvr* > *heelAvrgStandDetection* **then**
      Activity ← Toe Standing;
    else
      Activity ← Toe Sitting;
    end
  else
    if *HeelAvr* > *heelAvrgStandDetection* **then**
      Activity ← Normal Standing;
    else
      Activity ← Normal Sitting;
    end
  end
end

$$WindowAverage_t = \frac{\sum_{i=1}^{n} x_i}{n} \tag{1}$$

Where x_i is a signal reading and n is the size of the window which here is assumed to be 8.

$$\text{WindowStandardDeviation}_t = \left(\frac{\sum_{i=1}^{n}(x_i - WindowAverage_t)^2}{n-1}\right)^{\frac{1}{2}} \tag{2}$$

The same as equation (1), x_i is a signal reading and n is the size of the window.

Table I. lists the parameters that we employed in our detection algorithm in which toe and heel windows refer to the chopped

TABLE I
ACTIVITY RECOGNITION ALGORITHM PARAMETERS

Parameters	**Definitions**
toeStdWalkDetection	Set parameter for walk recognition regarding standard deviation of toe window
toeAvrgToeWalkDetection	Set parameter for toe walk recognition regarding average of toe window
heelAvrgToeWalkDetection	Set parameter for toe walk recognition regarding average of heel window
toeAvrgToeSitStandDetection	Set parameter for toe sit and stand recognition regarding average of toe window
heelAvrgStandDetection	Set parameter for stand recognition regarding average of heel window

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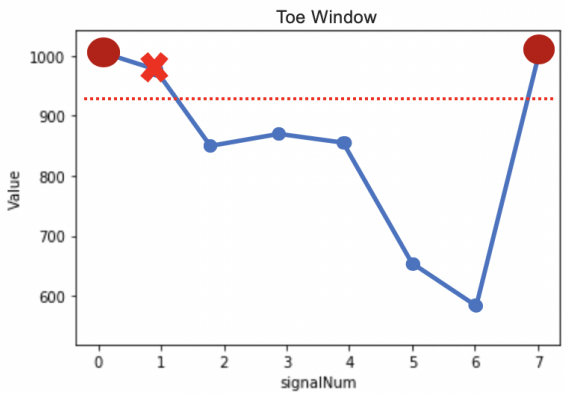
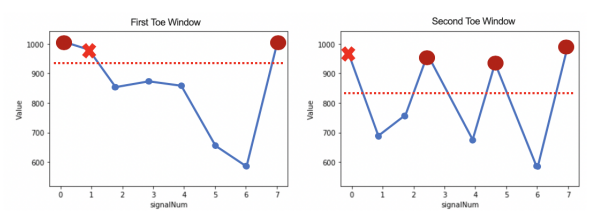


(a)



(b)

Fig. 3. Peak detection: (a) Detecting peaks in a window (b) Detecting peaks in two successive windows. The red dotted line passes through the average value of the eight signals. Signals above the red dotted line and highlighted by a red circle are counted as a peak (step). Signals highlighted by a red cross are above the average line but are not counted as a step.

8 captured signals of the FSRs located on the insole's toe and heel zone respectively. These parameters are set heuristically. Activity patterns are identified according to Algorithm1. Conditions for toe walking, sitting, and standing are characterized based on the definition of the toe walking that heel never touches the floor through the gait cycle. If activity is specified as normal or toe walking then the steps counting procedure takes place (Fig2). Regarding the gait cycle, toe or ball of the foot is off the ground during the swing phase which happens once for each step. Having this in mind, collecting peak signals of the toe window is an accurate estimation of step counts. To this end, for each toe window that is identified as a walking activity, the signal retaining greater value than the window average is classified as a peak of the window.

Taking into account various walking paces, consecutive peak points are considered as a single step. In a window containing 8 consecutive captured signals of the FSR on the insole's toe demonstrated in Fig3(a), signals indexed 0, 1, and 7 hold values higher than the average value of the eight signals in the window. The average of the signals in a window is indicated as a red dotted line in the Fig3. Due to the sequence of signals positioned 0 and 1 in Fig3(a), signal in position 1 is not labeled as a peak (step). Therefore, in Fig3(a), signals indexed 0 and 7 are counted as a step and marked by a red circle, however, the signal indexed 1 is not counted as a step and is marked by a red cross. Accordingly windows of toe sensor signals are processed one after another. As an example of handling consecutive windows in Fig3(b), the last signal (index 7) of first window is marked as a peak (step) and highlighted by a red circle. Incidentally, the first signal (index 0) of the second window carries larger value than the mean signal values of the second window. Since the first window's last signal and the second window's first signal are consecutive, the last one is not perceived as a peak (step) and is marked by a red cross.

As an intervention for the user, after each third, sixth, and ninth successive toe step detection, a short vibration with duration of a second is generated and there is a long vibration with duration of 30 seconds after the tenth successive toe step detection. The long vibration is halted as soon as a heel strike is perceived. The same intervention procedure takes place for being on the toes while sitting or standing. In this case, each 400ms sitting or standing on toes is taken in to account as a step unit for the procedure. We could verify that continuous usage of these interventions enhances subjects' gait pattern over time.

C. Mobile Application

Our Android application provides real-time activity monitoring and transfers data to the cloud-based database. A stacked bar chart on the application displays the number of toe steps in red and the number of normal heel to toe steps in green (Fig4). The application receives sensor readings and activity states from the BLE connected Smart Insole. The rightmost

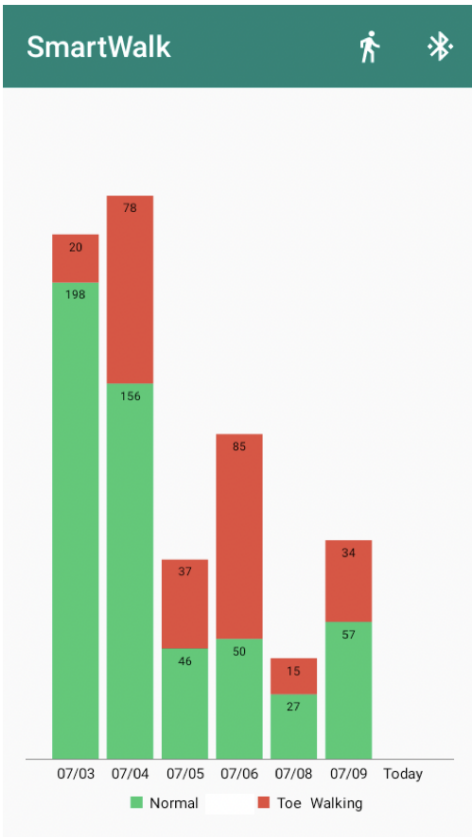


Fig. 4. Mobile application activity recognition 7-day bar chart

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TABLE II
EVALUATION PROTOCOL AND ACCURACY PERCENTAGE OF EACH ACTIVITY

Activity \ Subject Number	Accuracy (%)									
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
100-step regular pace normal walking	92	93	71	81	99	98	97	88	73	93
100-step regular pace toe walking	96	97	92	94	99	94	95	86	99	91
100-step slow pace normal walking	87	92	91	86	99	97	100	94	83	90
100-step slow pace toe walking	90	99	82	99	100	95	90	95	99	88
100-step fast pace normal walking	96	82	79	92	84	81	91	94	75	86
100-step fast pace toe walking	91	99	97	96	99	99	96	83	82	92
Average	92 ± 3.21	93.67 ± 5.88	85.34 ± 8.84	91.34 ± 6.10	96.67 ± 5.68	94 ± 6.06	94.83 ± 3.44	90 ± 4.58	85.17 ± 10.40	90 ± 2.38

bar-chart shows real-time step counts.The application also retrieves records from the database to exhibit the past 6-day of walking activity. Google Activity Recognition API was used to detect users' activity through the Android device and compare activity states from the Smart Insole. Our design of the app is not intrusive and minimizes user interaction to focus on reviewing the users' walking patterns and progress.

The application sends collected data to the Google Firebase via Wi-Fi. Firebase is a cloud-based NoSQL document database that permits the Android applications to access the database in real-time. The Firebase includes machine learning and data science analytics library, which will be useful for our further study with the collected data. It also provides real-time visualization of the data.

IV. EXPERIMENTS

We placed the compact system embedded in the insoles of U.S. adult size 6 and 11. To evaluate our activity recognition algorithm we did a study on 10 participants. Table II indicates the activities completed by each participant and the accuracy of each activity. First, we asked the participants to walk with their normal pace, then to adjust their pace to slow and fast relative to their normal pace. The average steps count accuracy of our algorithm is $91.3\% \pm 3.62$ various from a subject to another regarding their gait cycle. Our experiments show that the activity recognition algorithm works precisely for different range of walking paces. The participants performed each activity of toe sitting, toe standing, normal sitting, and normal standing for a minute of duration. As we expected, no step was detected during these experiments which delivers 100% accuracy for sitting and standing detection.

V. CONCLUSIONS

In this paper, we introduce Smart Insole, a remote-monitoring system for activity recognition and improving walking patterns of individuals that suffer from toe walking. Smart Insole provides a wearable, portable, reliable, and low cost platform to monitor walking patterns. We validated our proposed activity recognition and pedometer with an astute evaluation protocol. Our system is going to be examined by clinical patients and the experiment outcome can provide us beneficial information regarding the impact of our toe walking intervention technique on their walking patterns. We are actively enhancing our machine learning analytics component which can provide us additional helpful information. Smart Insole can benefit children walking on their toes to improve their walking patterns and orthopedic experts to monitor patients' activities and progress.

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